Mathematics GRE Subject Test

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January 31, 2025

Outline

1. Overview
   - What is the Mathematics GRE
2. Problems
   - Easy problems
   - Medium problems

Outline

Who,Where,When

- required by many applications to grad school
- optional for other applications
- take online at home or at ETS testing center
- 2024-2025 testing periods: 9/16/24 - 9/29/24, 10/17/24-10/30/24, and 04/21/25-05/04/25.
- can take it once during each "testing period"
- costs $150 (plus $250 for general exam)

Test content

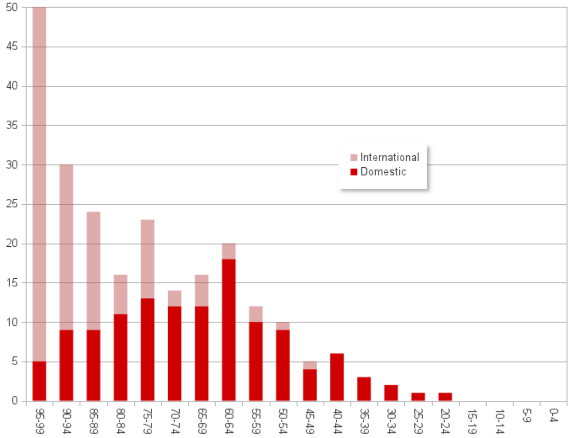
Exam basics

- time length: 2 hours 50 minutes
- number of problems: 66

Problem breakdown

- 50% calculus
- 25% elementary algebra, linear algebra, abstract algebra, number theory
- 25% misc. other topics: topology, complex numbers, graph theory, combinatorics

Foreign vs. domestic



Percentiles

Score	Percentile
960	96
880	90
840	82
780	70
740	62
680	50
640	42
580	30

Resources

- Princeton Review book
- 5 official practice tests (found online)
- `https://sites.math.rutgers.edu/~iacoley/greprep.html`
- `https://www.varsitytutors.com/gre_subject_test_math-practice-tests`
- google around!

Strategy

- The first 10-15 problems should be fast / easy
- Try to do easiest problems in your head!
- Harder problems: when we have to put pen to paper (toward middle)
- Harder-er problems: linear algebra, abstract algebra (mixed in everywhere)
- Hardest problems: special topics (towards the back)

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Problem 1

$$\lim_{x \to 0} \frac{\sin(4x) - 4x}{x^3}$$

Problem 2

Calculate

$$\int_{e^2}^{e^5} \frac{1}{x \log^2 x}$$

Problem 3

What is the value of the sum

$$\sum_{n=1}^{\infty} \frac{n^2+2}{n^3+3n+5}$$

Problem 4

For which value of a is

$$\int_a^{a+1} (x^2 + x)dx$$

a minimum?

Problem 5

If V and W are 5-dimensional subspaces of $\mathbb{R}^7$, what can the dimension of $V \cap W$ be?

Problem 6

What is the greatest possible area of a triangular region with one vertex at the center of a circle of radius 1 and the other two vertices on the circle?

Outline

Problem 1

Compute the second derivative of

$$x^{x^x}$$

Problem 2

Compute

$$\lim_{n\to\infty} \frac{5}{n} \sum_{i=1}^{n} \left[\left(\frac{2i}{n}\right)^5 - \left(\frac{2i}{n}\right)^3 \right]$$